

Proposition 5

Discrete check: when do low-knowledge workers gain from autonomous AI?

Consider a low-knowledge worker with knowledge l , initially matched with a human solver of knowledge $s > l$

A worker fails with probability $1-l$, and each failed problem uses h units of solvetime. Define

$$q \equiv h(1-l) \quad n(l) = \frac{1}{q}$$

Let w_H be the wage of the human solver

1. Before AI

Zero profit for a human team implies

$$\begin{aligned} n(l)[s - w_L] - w_H &= 0 \\ \frac{1}{q}[s - w_L] &= w_H \end{aligned}$$

so

$$w_L = s - qw_H$$

Thus the worker's wage depends positively on the solver's knowledge s , but the worker must indirectly compensate the solver through qw_H

2. Autonomous AI as solver

Now introduce autonomous AI with capability

$$a \equiv z_{AI}, \quad l < a < s$$

Because autonomous AI can also produce independently, its competitive rental rate is

$$r = a$$

If AI becomes the solver, zero profit gives

$$\begin{aligned} n(l)[a - w_L^A] - a &= 0 \\ \frac{1}{q}[a - w_L^A] - a &= 0 \end{aligned}$$

Therefore,

$$\frac{1}{q}[a - w_L^A] = a$$

and

$$w_L^A = a(1-q)$$

3. Wage effect on the bottom worker

Define

$$\Delta_L = w_L^A - w_L$$

Then

$$\Delta_L = a(1-q) - (s - q w_H)$$

$$\Delta_L = \underbrace{(a-s)}_{\text{match effect}} + \underbrace{q(w_H-a)}_{\text{Share effect}}$$

Since $a < s$

$$a - s < 0,$$

So replacing the human solver with AI worsens the worker's match quality

But, if $w_H > a$

$$q(w_H - a) > 0,$$

because AI is a cheaper solver. The worker therefore keeps a larger share of team output

Hence the worker gains only if

$$\text{share effect} > |\text{match effect}|$$

4. Capability cutoff

Set

$$\Delta_L = 0,$$

Then

$$a - s + q(w_H - a) = 0,$$

so

$$a(1-q) = s - q w_H$$

Therefore the cutoff is

$$\bar{a} = \frac{s - q w_H}{1 - q}$$

Since $1 - q > 0$

$$w_L^A > w_L \Leftrightarrow a > \bar{a}$$

So, holding autonomy fixed, low-knowledge workers benefit only when AI capability is sufficiently high

5. Numerical check

Take

$$S = 0.8, \quad W_H = 1, \quad q = 0.4$$

Then

$$w_L = 0.8 - 0.4(1) = 0.4,$$

and

$$\bar{a} = \frac{0.8 - 0.4}{0.6} = \frac{2}{3} \approx 0.667$$

If $a = 0.60 < \bar{a}$,

$$w_L^A = 0.60(0.60) = 0.36 < 0.4$$

$\Rightarrow$ Bottom worker loses

If $a = 0.70 > \bar{a}$,

$$w_L^A = 0.70(0.60) = 0.42 > 0.40$$

$\Rightarrow$ Bottom worker gains

Verdict

$$\text{Bottom gains} \Leftrightarrow Z_{AI} > \bar{Z}_{AI}$$

In this discrete analogue, the threshold comes from the positive share effect eventually dominating the negative match effect.

The key condition is about AI capability, not autonomy, because autonomy is held fixed throughout this comparison.